INCH-POUND

MS27789D
13 September 2005

SUPERSEDING
MS27789C
28 September 1987

DETAIL SPECIFICATION SHEET

SWITCH, TOGGLE, FOUR POLE, ENVIRONMENTALLY SEALED, LEVER LOCK, INTEGRATED WIRE TERMINALS

INACTIVE FOR NEW DESIGN AFTER 30 AUGUST 2005

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switch described herein shall consist of this specification and the latest issue of MIL-DTL-3950.

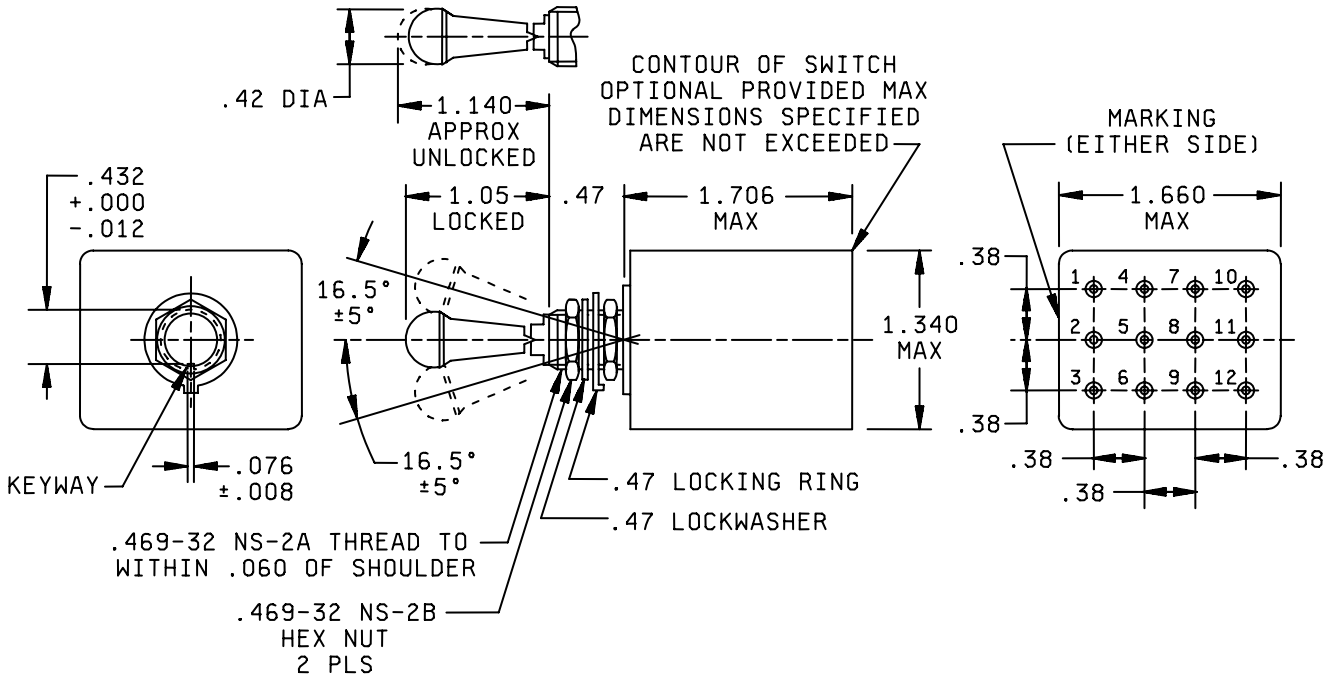


FIGURE1. Dimensions and configuration

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LOCKING COMBINATIONS

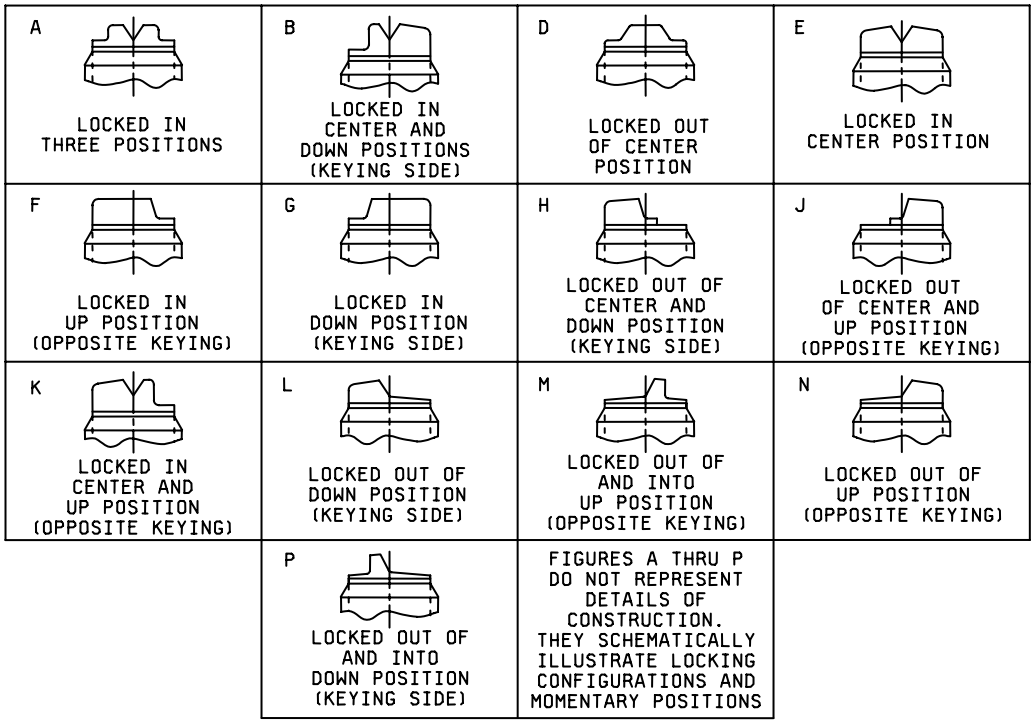


FIGURE 1 Dimensions and configuration- Continued

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Inches	mm	Inches	mm
.005	0.13	.432	10.97
.008	0.20	.469	11.91
.012	0.30	.47	11.9
.020	0.51	1.05	26.7
.060	1.52	1.140	28.96
.076	1.93	1.340	34.04
.38	9.7	1.660	42.16
.42	10.7	1.706	43.33

NOTES:

1. Dimensions are inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .020 (0.51 mm) for two place decimals and ± .005 (0.13 mm) for three place decimals.
4. For hardware detail specifications see appendix of MIL-DTL-3950.
5. Contact installing/removal tool in accordance with MIL-I-81969.
6. Grommet sealing plug in accordance with MS27488A20.
7. Sealing plugs may be used in nonfunctional grommet holes.
8. Sealing grommet shall seal on a smooth wire insulation of .040 (1.02 mm) to .083 (2.11 mm)
9. Terminal shall adequately accept a wire contact within dimensional limits of SAE-AS39029.
10. The terminal end of switch shall be color coded red to indicate contact size.
11. Direction of internal mechanism movement is opposite to the direction of the toggle movement.
12. Part number example: MS27789-21A (locking combination ‘A’)
13. In the event of a conflict between the text of this standard and the reference cited herein, the text of this standard shall take precedence.

FIGURE1. Dimensions and configuration- Continued

TEST REQUIREMENTS:

Test shall be performed in accordance with MIL-DTL-3950 except:

During all tests, switch shall be fully wired with appropriate wire and terminal contacts.

Contact voltage drop- the contact voltage drop with two terminals and the switch contact in series shall not exceed 8 millivolts measured from one wire contact through the contacts to the other wire contacts.

DETAIL REQUIREMENTS:

Locking arrangement: Positive locking shall be accomplished and shall prevent motion of the toggle lever until the locking mechanism is manually released.

The force required to release the locking mechanism shall be 3 to 5 pounds.

The locking means at the top of the toggle bushing shall be capable of withstanding a torque of 20 inch-pounds applied in both directions immediately following the humidity test.

Weight: .227 pound maximum.

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MS part No.	Available locking combinations	Circuit with Toggle Lever In			Current capacity (amperes) 28 volts DC			Current capacity(amperes) 115 volts, 60 and 400 Hertz AC		
		Keying side	center	Opposite keying side	Lamp-Load circuit	Resistive circuit	Inductive circuit	Lamp-Load circuit	Resistive circuit	Inductive circuit
MS27789-1	ALL	1-2 4-5 7-8 on 10-11	2-3 4-5 7-8 on 11-12	2-3 5-6 8-9 On 11-12	5	7.5	7.5	4	7.5	7.5
MS27789-2	E, F, K, L, M, N	1-2 Mom. 4-5 7-8 on 10-11	2-3 4-5 7-8 on 11-12	2-3 5-6 8-9 on 11-12	4	7.5	7.5	2	7.5	7.5
MS27789-3	E, L, N	1-2 Mom. 4-5 7-8 on 10-11	2-3 4-5 7-8 on 11-12	2-3 Mom. 5-6 8-9 on 11-12						
MS27789-21	ALL	1-2 7-8 4-5 on 10-11	Off	2-3 8-9 8-9 on 11-12	5	7.5	7.5	4	7.5	7.5
MS27789-22	D, F, G	Off	None	2-3 8-9 5-6 on 11-12						
MS27789-23	D, F, G	1-2 Mom. 7-8 4-5 on 10-11	None	2-3 8-9 5-6 on 11-12						
MS27789-24	E, F, K, M	None	Off	2-3 8-9 5-6 on 11-12						
MS27789-25	F	None	Mom. Off	2-3 8-9 5-6 on 11-12	4	7.5	7.5	2	7.5	7.5
MS27789-26	F	1-2 Mom. 7-8 4-5 on 10-11	None	2-3 8-9 5-6 on 11-12						
MS27789-27	E, L, N	1-2 Mom. 7-8 4-5 on 10-11	Off	2-3 8-9 5-6 on 11-12						
MS27789-28	E	1-2 Mom. 7-8 4-5 on 10-11	Off	None						
MS27789-29	F	Mom. Off	None	2-3 8-9 5-6 on 11-12						
MS27789-30	F	1-2 Mom. 7-8 4-5 on 10-11	None	Off						
MS27789-31	E, F, K, L, M, N	1-2 Mom. 7-8 4-5 on 10-11	off	2-3 8-9 5-6 on 11-12						
MS27789-32	E	None	1-2 7-8 4-5 on 10-11	2-3 Mom. 8-9 5-6 on 11-12						
MS27789-33	E, F, K, M	None	1-2 7-8 4-5 On 10-11	2-3 8-9 5-6 on 11-12	5	7.5	7.5	4	7.5	7.5

Referenced documents
MIL-DTL-3950
MIL-I-81969
MS27488
SAE-AS39029

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – CR
Navy - AS
Air Force - 11
DLA – CC

Preparing activity
DLA -CC
(Project 5930-2005-003)

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Review activities
Army – AR, AV, CR4
Navy – EC, MC
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/

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